Lesson 12: Some Important Discrete RVs

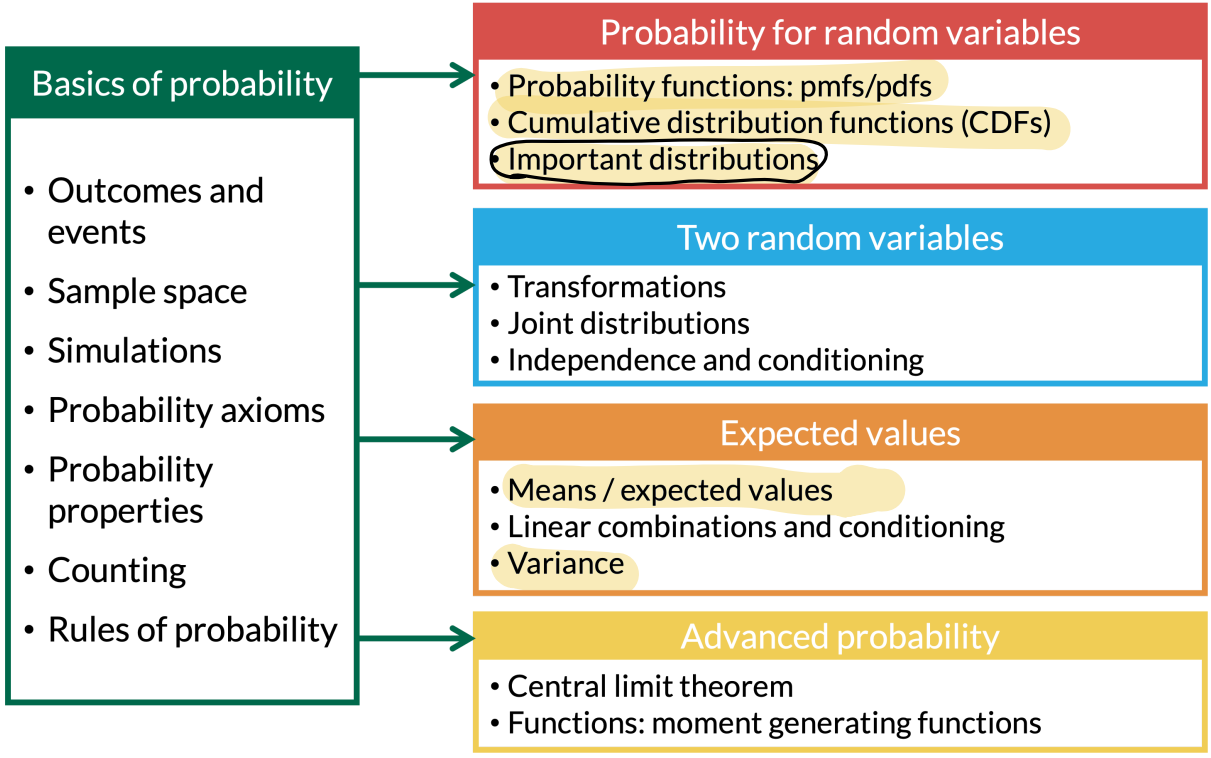
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Learning Objectives

1. Distinguish between Bernoulli, Binomial, Geometric, Hypergeometric, Discrete Uniform, Negative Binomial, and Poisson distributions when reading a word problem.
2. Identify the variable and the parameters in a word problem, and state what the variable and parameters mean.
3. Use the formulas for the pmf/CDF, expected value, and variance to answer questions and find probabilities.

Where are we?


Basics of probability
• Outcomes and events
• Sample space
• Simulations
• Probability axioms
• Probability properties
• Counting
• Rules of probability
Probability for random variables
• Probability functions: pmfs/pdfs
• Cumulative distribution functions (CDFs)
• Important distributions
Two random variables
• Transformations
• Joint distributions
• Independence and conditioning
Expected values
• Means / expected values
• Linear combinations and conditioning
• Variance
Advanced probability
• Central limit theorem
• Functions: moment generating functions

Bernoulli RVs

Properties of Bernoulli RVs

- **Scenario:** One trial, with outcome success or failure
- Shorthand: $X \sim \text{Bernoulli}(p)$

X

$$X = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1-p \end{cases}$$

success

failure

pmf

$$p_X(x) = P(X = x) = p^x (1-p)^{1-x} \text{ for } x = 0, 1$$

$\mathrm{E}(X)$

$$\mathrm{E}(X) = p$$

$\mathrm{Var}(X)$

$$\mathrm{Var}(X) = pq = p(1-p) \quad \text{where } q = 1-p$$

Bernoulli RVs we've already seen

- We've seen a few examples of Bernoulli distributions:
  - **Lesson 1:** Toss one coin. Success is heads, so $p = \frac{1}{2}$
  - **Lesson 7, Example 1 (falls in older adults):** Whether one older adult has at least one fall in a year, $p = 0.25$
  - **Lesson 10, Example 3:** The general Bernoulli RV, where we found $\mathrm{E}(X) = p$
  - **Lesson 11, Example 2 (card example):** Draw one card, and $X = 1$ if it is a heart, $p = \frac{1}{4}$
  - **Lesson 11, Example 5 (ghost problem):** The ghost grabs one candy, and $Y = 1$ if it is a chocolate, $p = \frac{10}{60} = \frac{1}{6}$

Binomial RVs

Properties of Binomial RVs

- **Scenario:** There are n independent trials, each resulting in a success or failure, with constant probability, p, in each trial. We are counting the number of successes (or failures).
- Shorthand: $X \sim \text{Binomial}(n, p)$
- A **binomial random variable** is the **sum of n independent Bernoulli RVs**

X	$X = \text{Number of successes of } n \text{ independent trials}$
pmf	$p_X(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$ for $x = 0, 1, 2, ..., n$
$\text{E}(X)$	$\text{E}(X) = np$
$\text{Var}(X)$	$\text{Var}(X) = npq = np(1-p)$

CDF: $P(X \leq \quad) = \sum_{i=1}^{k} P(X = x)$

Ghost problem revisited

Example 1

Recall the ghost problem from Lesson 11 (Example 5). The bag of candy has 10 chocolates, 20 lollipops, and 30 laffy taffies. The ghost grabs a handful of five pieces of candy ***with*** replacement.

- Let Y be the number of chocolates the ghost takes.
- What is the distribution of Y?
- Find the mean and variance of Y.

$n = 5 \quad p = \frac{10}{60} = \frac{1}{6}$

$Y \sim \text{Binomial}\left(n = 5, p = \frac{1}{6}\right)$

$E(Y) = E\left(\sum_{i=1}^{5} X_i\right)$

↑ Bernoulli

$E(Y) = np = 5\left(\frac{1}{6}\right) = \frac{5}{6}$

$\text{Var}(Y) = np(1-p) = 5\left(\frac{1}{6}\right)\left(1-\frac{1}{6}\right)$

$= 5 \cdot \frac{1}{6} \cdot \frac{5}{6} = \frac{25}{36}$

Binomial distribution: R commands

Recap from Lesson 7. R commands with their input and output:

R code	What does it return?	What do we use it for?
`rbinom()`	returns sample of random variables with specified binomial distribution	Simulate a Binomial RV
`dbinom()`	returns probability of getting certain number of successes	$P(X = x)$
`pbinom()`	returns cumulative probability of getting certain number or less successes	$P(X \leq x)$ (or $P(X > x)$ if we add `lower.tail = FALSE`)
`qbinom()`	returns number of successes corresponding to desired quantile (p) → k	Goes the other direction: we give it a probability p and it returns the value k where $P(X \leq k) = p$

- A Bernoulli RV is a binomial RV with $n = 1$, so we use the same commands with `size = 1`

```
1 dbinom(1, size = 1, prob = 10/60)
```

```
[1] 0.1666667
```

Ghost problem revisited: simulate it

Example 1

Simulate Y, the number of chocolates in 5 pieces, 10,000 times. Do the mean and variance match $np = \frac{5}{6} \approx 0.833$ and $np(1-p) = \frac{25}{36} \approx 0.694$?

```
1 set.seed(1104)                                          ①
2 reps <- 10000                                           ②
3 n_candy <- 5                                            ③
4 p <- 10 / 60
5
6 ghost_sims <- tibble(sim = 1:reps) |>                   ④
7   mutate(
8     Y = rbinom(n(), size = n_candy, prob = p)           ⑤
9     )
```

repititions of the binomial

n in our pmf / E(X) / Var(X)

```
1 ghost_sims |>                                           ①
2   summarise(
3     average  = mean(Y),                                 ②
4     variance = mean((Y - mean(Y))^2),                   ③
5     prob_1   = mean(Y == 1)                             ④
6   )
```

```
# A tibble: 1 × 3
  average variance prob_1
    <dbl>    <dbl>  <dbl>
1   0.838    0.700  0.394
```

Connecting to Lesson 11

In Lesson 11 we found the variance by adding up 5 Bernoulli variances. Now the binomial formula does it in one step!

Geometric RVs

Geometric RVs

- **Scenario:** There are repeated independent trials, each resulting in a success or failure, with constant probability of success for each trial. We are counting the number of trials until the first success.
  - For example, toss a coin until we get heads: H, or TH, or TTH, or TTTH, ...
- Shorthand: $X \sim \text{Geo}(p)$ or $X \sim \text{Geometric}(p)$ or $X \sim \text{G}(p)$

	X	X
	Number of trials needed for first success (count x *includes* the success)	**Number of failures before first success** (count x *does not include* the success)
pmf	$p_X(x) = P(X = x) = (1-p)^{x-1}p$ for $x = 1, 2, 3, \ldots$	$p_X(x) = P(X = x) = (1-p)^{x}p$ for $x = 0, 1, 2, \ldots$
CDF	$F_X(x) = P(X \leq x) = 1 - (1-p)^{x}$ for $x = 1, 2, 3, \ldots$	$F_X(x) = P(X \leq x) = 1 - (1-p)^{x+1}$ for $x = 0, 1, 2, \ldots$
E(X)	$E(X) = \frac{1}{p}$	$E(X) = \frac{1-p}{p}$
Var(X)	$Var(X) = \frac{1-p}{p^2}$	$Var(X) = \frac{1-p}{p^2}$

Bullseye (1/4)

Example 2

We throw darts at a dartboard until we hit the bullseye. Assume throws are independent and the probability of hitting the bullseye is 0.01 for each throw.

1. What is the pmf for the number of throws needed to hit the bullseye? # trials (not failures)
2. What are the mean and variance for the number of throws needed to hit the bullseye?
3. Find the probability that our first bullseye:
   - is on one of the first fifty tries
   - is after the 50th try, given that it did not happen on the first 20 tries

Bullseye (2/4)

Example 2

We throw darts at a dartboard until we hit the bullseye. Assume throws are independent and the probability of hitting the bullseye is 0.01 for each throw.

1. What is the pmf for the number of throws needed to hit the bullseye?

- Let X = number of throws needed to hit the first bullseye (the count *includes* the bullseye)
- Possible sequences, with H = hit and M = miss:
  - H ($x = 1$), MH ($x = 2$), MMH ($x = 3$), MMMH ($x = 4$), ...
- Each throw is independent, so $x - 1$ misses and then 1 hit:

$$P(X = x) = (1 - p)^{x-1} p^1$$

- With $p = 0.01$:

$$p_X(x) = 0.99^{x-1}(0.01) \quad \text{for } x = 1, 2, 3, \ldots$$

- $X \sim \text{Geo}(0.01)$

Bullseye (3/4)

$$E(X) = \frac{1}{p} \qquad Var(X) = \frac{1-p}{p^2}$$

Example 2

We throw darts at a dartboard until we hit the bullseye. Assume throws are independent and the probability of hitting the bullseye is 0.01 for each throw.

2. What are the mean and variance for the number of throws needed to hit the bullseye?

$$E(X) = \frac{1}{0.01} = 100$$

$$Var(X) = \frac{1-0.01}{(0.01)^2} = \frac{0.99}{0.0001}$$

$$= 9,900$$

Bullseye (4/4)

Example 2

We throw darts at a dartboard until we hit the bullseye. Assume throws are independent and the probability of hitting the bullseye is 0.01 for each throw.

3. Find the probability that our first bullseye:

(a) • is on one of the first fifty tries

(b) • is after the 50th try, given that it did not happen on the first 20 tries

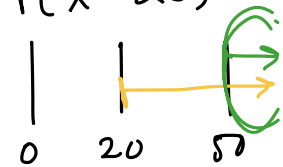
(a) $P(X \leq 50) = F_X(x = 50)$

$= 1 - (1 - 0.01)^{50}$

$= 1 - 0.99^{50} = 0.395$

(b) $P(\underbrace{X > 50}_{A} \mid \underbrace{X > 20}_{B}) = \frac{P(X > 50, X > 20)}{P(X > 20)}$

↳ $P(A|B) = \frac{P(A \cap B)}{P(B)}$



$= \frac{P(X > 50)}{P(X > 20)} = \frac{1 - F_X(50)}{1 - F_X(20)}$

$= \frac{1 - [1 - (1 - 0.01)^{50}]}{1 - [1 - (1 - 0.01)^{20}]}$

$= \frac{0.99^{50}}{0.99^{20}} = 0.99^{30}$ ← $1 - F_X(30)$

Memoryless property for Geometric RVs

If we know X is greater than some number (aka given $X > j$), then the probability of $X > k + j$ is just the probability that $X > k$.

$$P(X > 50 \mid X > 20) = P(X > 30)$$

$k+j$ $\quad$ j $\quad$ k

This is the **memoryless** property.

$$P(X > k + j | X > j) = P(X > k)$$

$(1-p)^k(1-p)^j$

$$P(X > k + j | X > j) = \frac{P(X > k + j \text{ and } X > j)}{P(X > j)} = \frac{P(X > k + j)}{P(X > j)} = \frac{(1-p)^{k+j}}{(1-p)^j} = (1-p)^k$$

Geometric distribution: R commands

R commands with their input and output:

R code	What does it return?	What do we use it for?
`rgeom()`	returns sample of random variables with specified geometric distribution	Simulate a Geometric RV
`dgeom()`	returns probability of getting certain number of failures before the first success	$P(X = x)$
`pgeom()`	returns cumulative probability of getting certain number of failures or less	$P(X \leq x)$ (or $P(X > x)$ if we add `lower.tail = FALSE`)
`qgeom()`	returns number of failures corresponding to desired quantile	Goes the other direction: we give it a probability and it returns the value k where $P(X \leq k)$ reaches it

- **Careful:** R counts the **failures** before the first success (the right column of our table). If X counts the **trials**, use $x - 1$:

```
1 pgeom(50 - 1, prob = 0.01)
[1] 0.3949939
```

$P(X \leq 50)$ ← trials

Let's simulate the bullseye example (1/2)

- `rgeom()` simulates a geometric RV directly, so we don't need to act out every throw

```
1 p <- 0.01                                                    ①
2
3 bullseye_sims <- tibble(sim = 1:reps) |>                     ②
4   mutate(
5     misses = rgeom(n(), prob = p),                           ③
6     X = misses + 1                                           ④
7   )
```

- The first few simulated values:

```
1 bullseye_sims                                                ①
# A tibble: 10,000 × 3
     sim misses     X
   <int>  <int> <dbl>
 1     1      4     5
 2     2     17    18
 3     3     65    66
 4     4    223   224
 5     5    121   122
 6     6     33    34
 7     7    120   121
 8     8    112   113
 9     9      7     8
10    10      4     5
# ℹ 9,990 more rows
```

Let's simulate the bullseye example (2/2)

```
1 bullseye_sims |>                                    ①
2   summarise(
3     average       = mean(X),                       ②
4     variance      = mean((X - mean(X))^2),
5     prob_first_50 = mean(X <= 50)                  ③
6   )
```

```
# A tibble: 1 × 3
  average variance prob_first_50
    <dbl>    <dbl>         <dbl>
1    100.   10012.         0.395
```

- By hand: $P(X \leq 50) = 1 - 0.99^{50} \approx 0.395$

- Check the memoryless property

```
1 bullseye_sims |>                                    ①
2   filter(X > 20) |>
3   summarise(prob_after_50 = mean(X > 50))           ②
4
5 bullseye_sims |>                                    ③
6   summarise(prob_after_30 = mean(X > 30))
```

```
# A tibble: 1 × 1
  prob_after_50
          <dbl>
1         0.740
# A tibble: 1 × 1
  prob_after_30
          <dbl>
1          0.74
```

- Both are close to $0.99^{30} \approx 0.740$

Negative Binomial RVs

Properties of Negative Binomial RVs

- **Scenario:** There are repeated independent trials, each resulting in a success or failure, with constant probability of success for each trial. We are counting the number of trials until the rth success.
- Shorthand: $X \sim \text{NegBin}(p, r)$ or $X \sim \text{NB}(p, r)$
- Negative binomial is sum of r geometric distributions

$r = 2$

S S
F S S
F S F S
F F S S

X (includes successes) $X = \text{Number of independent trials until } r^{th} \text{ success}$ through

pmf $$p_X(x) = P(X = x) = \binom{x-1}{r-1}(1-p)^{x-r}p^r \text{ for } x = r, r+1, r+2, \ldots$$

#failures

r = # successes

E(X) $$E(X) = \frac{r}{p}$$

$Y \sim G(p)$

$E(X) = E\left(\sum_{i=1}^{r} Y\right) = r E(Y)$

Var(X) $$Var(X) = \frac{rq}{p^2} = \frac{r(1-p)}{p^2}$$

$\text{Var}(X) = r\,\text{Var}(Y)$

Hitting more than 1 bullseye

trials including successes

Example 3

Consider again the bullseye example, where we throw darts at a dartboard until we hit the bullseye. Assume throws are independent and the probability of hitting the bullseye is 0.01 for each throw.

1. What is the expected value and variance of the number of throws needed to hit 5 bullseyes?

- Let X = number of throws needed to hit 5 bullseyes
- Parameters: $r = 5$ bullseyes and $p = 0.01$
- $X \sim \text{NegBin}(r = 5, p = 0.01)$

$$E(X) = \frac{r}{p} = \frac{5}{0.01} = 500$$

5 · 100

$$Var(X) = \frac{r(1-p)}{p^2} = \frac{5(0.99)}{0.01^2} = \frac{4.95}{0.0001} = 49{,}500$$

5 · 9,900

- Check with the geometric: one bullseye takes $E(Y) = 100$ throws with $Var(Y) = 9{,}900$, so 5 bullseyes take $5 \cdot 100 = 500$ throws with variance $5 \cdot 9{,}900 = 49{,}500$

Hitting more than 1 bullseye

X = # throws to get 5 bullseyes

$X \sim NB(r = 5, p = 0.01)$

Example 3

Consider again the bullseye example, where we throw darts at a dartboard until we hit the bullseye. Assume throws are independent and the probability of hitting the bullseye is 0.01 for each throw.

2. What is the probability that the 5th bullseye is on the 20th throw?

$$P(X = x) = \binom{x-1}{5-1}(1-p)^{x-5}p^5$$

$$P(X = 20) = \underbrace{\binom{20-1}{5-1}}_{\text{\# of ways to arrange 1st 19 throws}}(1-p)^{15}p^5$$

$$= 0.00000033$$

$$= 3.3 \times 10^{-7}$$

Negative binomial distribution: R commands

R commands with their input and output:

R code	What does it return?	What do we use it for?
`rnbinom()`	returns sample of random variables with specified negative binomial distribution	Simulate a Negative Binomial RV
`dnbinom()`	returns probability of getting certain number of failures before the rth success	$P(X = x)$
`pnbinom()`	returns cumulative probability of getting certain number of failures or less	$P(X \leq x)$ (or $P(X > x)$ if we add `lower.tail = FALSE`)
`qnbinom()`	returns number of failures corresponding to desired quantile	Goes the other direction: we give it a probability and it returns the value k where $P(X \leq k)$ reaches it

- **Careful:** like the geometric, R counts the **failures**. If X counts the **trials**, use $x - r$. `size` is r, the number of successes.

```
1 dnbinom(20 - 5, size = 5, prob = 0.01)
```

```
[1] 3.333586e-07
```

Hitting more than 1 bullseye: simulate it

Example 3

Simulate X, the number of throws needed to hit 5 bullseyes, 10,000 times. Do the mean and variance match $\frac{r}{p} = 500$ and $\frac{r(1-p)}{p^2} = 49500$?

```
1  r <- 5                                                          ①
2
3  bullseye5_sims <- tibble(sim = 1:reps) |>                       ②
4    mutate(
5      misses = rnbinom(n(), size = r, prob = p),                  ③
6      X = misses + r                                              ④
7    )
```

```
1  bullseye5_sims |>                                               ①
2    summarise(
3      average  = mean(X),                                         ②
4      variance = mean((X - mean(X))^2),
5      prob_20  = mean(X == 20)                                    ③
6    )
```

```
# A tibble: 1 × 3
  average variance prob_20
    <dbl>    <dbl>   <dbl>
1    504.   50303.       0
```

- `prob_20` is 0! $P(X = 20) \approx 0.00000033$ is so small that 10,000 simulations never see it. For very rare events, use `dnbinom()` instead.

3.3×10^{-7}

5 minute break

Poisson RVs

Properties of Poisson RVs

- **Scenario:** We are counting the number of successes in a **fixed time period (or fixed space)**, which has a constant **rate** (λ) of successes
- Shorthand: $X \sim \text{Poisson}(\lambda)$ or $X \sim \text{Pois}(\lambda)$

X	$X = \text{Number of successes in a given period (or given space)}$
pmf	$p_X(x) = P(X = x) = \dfrac{e^{-\lambda}\lambda^x}{x!}$ for $x = 0, 1, 2, 3, \ldots$
$\text{E}(X)$	$\text{E}(X) = \lambda$
$\text{Var}(X)$	$\text{Var}(X) = \lambda$

Distinguishing between Binomial and Poisson RVs

Binomial: $X \sim \text{Binomial}(n, p)$

- X models the number of successes ...
- in n independent (Bernoulli) trials ...
- that each have the same **probability** of success, p

Poisson: $X \sim \text{Poisson}(\lambda)$

- X models the number of successes ...
- in a **fixed time period** (or space), instead of n discrete independent trials ...
- that happen at a constant **rate**, λ, instead of a probability

Examples of Poisson RVs

- Number of visitors to an emergency room in an hour during a weekend night
- Number of study participants enrolled in a study per week
- Number of pedestrians walking through a square mile
- Number of babies born with a cleft palate within 1 year (rare events)
- Any more?

Emergency Room Visitors

Example 4

Suppose an emergency room has an average of 50 visitors per day. Find the following probabilities.

1. Probability of 30 visitors in a day
2. Probability of 8 visitors in an hour.

① $\lambda = 50$ visits/day

$X = \#$ visits (in a day)

$X \sim \text{Pois}(\lambda = 50)$

$$P(X = 30) = \frac{e^{-50} 50^{30}}{30!} = 0.000677$$

② $X = \#$ visits (in an hour)

$$\lambda = \frac{50 \text{ visits}}{1 \text{ day}} \cdot \frac{1 \text{ day}}{24 \text{ hours}} = \frac{50 \text{ visits}}{24 \text{ hrs}}$$

$$= 2.08\overline{3} \text{ visits/hr}$$

$$P(X = 8) = \frac{e^{-2.083} 2.083^{8}}{8!} = 0.00109$$

WRONG WAY: 8.24 visitors per day?

$P(X = 192)$ for $\lambda = 50$

RIGHT: ★ change λ, not X to fit original λ

Poisson distribution: R commands

R commands with their input and output:

R code	What does it return?	What do we use it for?
`rpois()`	returns sample of random variables with specified Poisson distribution	Simulate a Poisson RV
`dpois()`	returns probability of getting certain number of successes in the time period	$P(X = x)$
`ppois()`	returns cumulative probability of getting certain number or less successes	$P(X \leq x)$ (or $P(X > x)$ if we add `lower.tail = FALSE`)
`qpois()`	returns number of successes corresponding to desired quantile	Goes the other direction: we give it a probability and it returns the value k where $P(X \leq k)$ reaches it

- The only parameter is the rate, `lambda` $= \lambda$. Make sure it matches the time period in the question!

Emergency Room Visitors: in R

Example 4

Suppose an emergency room has an average of 50 visitors per day.

1. Probability of 30 visitors in a day.
2. Probability of 8 visitors in an hour.

- Probability of 30 visitors in a day.

```
1 dpois(30, lambda = 50)                                    ①
```

```
[1] 0.0006771985
```

- Probability of 8 visitors in an hour.

```
1 lambda_hour <- 50 / 24                                    ②
2 dpois(8, lambda = lambda_hour)                            ③
```

```
[1] 0.001095896
```

Change λ, not X

When the time period changes, change the **rate** λ to match it.

- Right way: $\lambda = \frac{50}{24}$ visitors per hour, and find $P(X = 8)$
- Wrong way: keep $\lambda = 50$ per day and find $P(X = 8 \cdot 24) = P(X = 192)$

Combining independent Poisson distributions

Theorem 1

If $X \sim Pois(\lambda_1)$ and $Y \sim Pois(\lambda_2)$ are independent of each other, then

$$Z = X + Y \sim Pois(\lambda_1 + \lambda_2)$$

Two emergency rooms

Example 5

Suppose emergency room 1 has an average of 50 visitors per day, and emergency room 2 has an average of 70 visitors per day, independently of each other. What is the probability distribution that models the total number of visitors to both?

- Let X = number of visitors to emergency room 1 in a day, so $X \sim \text{Pois}(\lambda_1 = 50)$
- Let Y = number of visitors to emergency room 2 in a day, so $Y \sim \text{Pois}(\lambda_2 = 70)$
- Let $Z = X + Y$ = total number of visitors to both in a day
- X and Y are independent, so by Theorem 1:

$$Z \sim \text{Pois}(\lambda_1 + \lambda_2) = \text{Pois}(50 + 70) = \text{Pois}(120)$$

$$p_Z(z) = P(Z = z) = \frac{e^{-120}\, 120^z}{z!} \quad \text{for } z = 0, 1, 2, \ldots$$

Two emergency rooms: simulate it

$E(Z) = \lambda = \lambda_1 + \lambda_2 = 50 + 70$

Example 5

Simulate one day of visitors at each emergency room 10,000 times. Does the total, $Z = X + Y$, have the mean and variance of a $\text{Pois}(120)$?

```
1 er_sims <- tibble(sim = 1:reps) |>                    ①
2   mutate(
3     X = rpois(n(), lambda = 50),                      ②
4     Y = rpois(n(), lambda = 70),                      ③
5     Z = X + Y                                         ④
6   )
```

λ_1 λ_2

```
1 er_sims |>                                            ①
2   summarise(
3     average  = mean(Z),                               ②
4     variance = mean((Z - mean(Z))^2)                  ③
5   )
```

```
# A tibble: 1 × 2
  average variance
    <dbl>    <dbl>
1     120.     120.
```

$Var(Z) = \lambda$

- For a Poisson RV, the mean and the variance are both λ
- Both are close to 120, just like Theorem 1 says!

Poisson Approximation of the Binomial

Both Poisson and Binomial RVs are counting the number of successes

- If for a Binomial RV
  - the number of trials n is very large, and
  - the probability of success p is small (close to 0),
- Then the Poisson distribution can be used to approximate Binomial probabilities
  - and we use $\lambda = np$ rare successes (events)
  - If p is close to 1, count the failures instead, with $\lambda = n(1-p)$
- **Rule of thumb:** We can use the Poisson approximation when $n \geq 20$ and $p \leq 0.05$, or when $n \geq 100$ and $np \leq 10$

Medical lab errors (at home reference)

Example 6

Suppose that in the long run, errors in a medical testing lab are made 0.1% of the time. Find the probability that fewer than 4 mistakes are made in the next 2,000 tests.

1. Find the probability using the Binomial distribution.
2. Approximate the probability in part (1) using the Poisson distribution.

Let X = number of mistakes in the next 2,000 tests

1. Binomial: $X \sim \text{Binomial}(n = 2000, p = 0.001)$

$$P(X < 4) = P(X \leq 3) = \sum_{x=0}^{3} \binom{2000}{x} (0.001)^x (0.999)^{2000-x} \approx 0.8572$$

- In R: `pbinom(3, size = 2000, prob = 0.001)`

2. Poisson approximation: $n = 2000 \geq 100$ and $np = 2 \leq 10$, so we can use $\lambda = np = 2$

$$P(X \leq 3) \approx \sum_{x=0}^{3} \frac{e^{-2} 2^x}{x!} = e^{-2} \left(1 + 2 + 2 + \frac{4}{3}\right) \approx 0.8571$$

- In R: `ppois(3, lambda = 2)`
- The two answers are very close!

Hypergeometric RVs

Hypergeometric RVs

probability NOT constant for every trial

- **Scenario:** There are a fixed number of successes and failures (which are known in advance), from which we make n draws without replacement. We are counting the number of successes from the n trials.
  - There is a finite population of N items
  - Each item in the population is either a success or a failure, and there are M successes total (and $N - M$ failures).
  - We randomly select (sample) n items from the population without replacement
- Shorthand: $X \sim \text{Hypergeo}(M, N, n)$

X

$$X = \text{Number of successes in } n \text{ draws}$$

pmf

$$p_X(x) = P(X = x) = \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}$$

for integer x with $\max(0, n - (N - M)) \leq x \leq \min(n, M)$

if $N = 100$, $N - M = 2$, $n = 5$ # of fail in "pop" N

min X is 3

$\text{E}(X)$

$$\text{E}(X) = \frac{nM}{N}$$

$\text{Var}(X)$

$$\text{Var}(X) = n\frac{M}{N}\left(1 - \frac{M}{N}\right)\left(\frac{N-n}{N-1}\right)$$

Wolf population

Example 7

A wildlife biologist is using mark-recapture to research a wolf population. Suppose a specific study region is known to have 24 wolves, of which 11 have already been tagged. If 5 wolves are randomly captured, what is the probability that 3 of them have already been tagged?

Let X = # of wolves tagged out of 5 (n)

$n = 5$ $\quad N - M = 13$

$N = 24$

$M = 11$

looking $P(X = 3)$

$$P(X = x) = \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}} = \frac{\binom{11}{x}\binom{13}{5-x}}{\binom{24}{5}}$$

$$P(X = 3) = \frac{\binom{11}{3}\binom{13}{2}}{\binom{24}{5}}$$

R command choose(top, bottom)

Hypergeometric distribution: R commands

R commands with their input and output:

R code	What does it return?	What do we use it for?
`rhyper()`	returns sample of random variables with specified hypergeometric distribution	Simulate a Hypergeometric RV
`dhyper()`	returns probability of getting certain number of successes in n draws	$P(X = x)$
`phyper()`	returns cumulative probability of getting certain number or less successes	$P(X \leq x)$ (or $P(X > x)$ if we add `lower.tail = FALSE`)
`qhyper()`	returns number of successes corresponding to desired quantile	Goes the other direction: we give it a probability and it returns the value k where $P(X \leq k)$ reaches it

- **Careful:** R uses different letters than we do. In R, `m` $= M$ (successes), `n` $= N - M$ (**failures**), and `k` $= n$ (draws): `dhyper(x, m = M, n = N - M, k = n)`

Wolf population: simulate it

- `rhyper()` captures 5 of the 24 wolves **without replacement** for us

```
1  M <- 11                                          ①
2  N <- 24
3  n_captured <- 5
4
5  wolf_sims <- tibble(sim = 1:reps) |>             ②
6    mutate(
7      X = rhyper(
8        n(),
9        m = M,
10       n = N - M,
11       k = n_captured
12       )                                          ③
13   )
```

```
1  wolf_sims |>                                     ①
2    summarise(
3      average = mean(X),                           ②
4      prob_3  = mean(X == 3)                       ③
5    )
```

```
# A tibble: 1 × 2
  average prob_3
    <dbl>  <dbl>
1    2.29  0.301
```

- The exact answer, two ways:

```
1  choose(11, 3) * choose(13, 2) / choose(24, 5)   ①
2  dhyper(3, m = 11, n = 13, k = 5)                ②
```

```
[1] 0.302795
[1] 0.302795
```

Binomial approximation of the hypergeometric RV

Suppose a hypergeometric RV X has the following properties:

- the population size N is really big,
- and the number of items n selected is small (relative to N)
- **Rule of thumb:** $\frac{n}{N} < 0.05$ (equivalently, $N > 20n$)

Then, in this case, making n draws from the population doesn't change the probability of success much, and the **hypergeometric RV can be approximated by a binomial RV** with $p = \frac{M}{N}$

- $\frac{M}{N}$ is like p in the binomial: the binomial has a constant p

Wolf population revisited (1/2)

Example 8

Suppose a specific study region is known to have 2400 wolves, of which 1100 have already been tagged.

1. If 50 wolves are randomly captured, what is the probability that 20 of them have already been tagged?

- Let X = number of tagged wolves in the 50 captured
- Parameters: $N = 2400$ wolves, $M = 1100$ tagged, $N - M = 1300$ untagged, and $n = 50$ captured
- $X \sim \text{Hypergeo}(M = 1100, N = 2400, n = 50)$

$$P(X = 20) = \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}} = \frac{\binom{1100}{20}\binom{1300}{30}}{\binom{2400}{50}} \approx 0.0813$$

- In R: `dhyper(20, m = 1100, n = 1300, k = 50)`

Wolf population revisited (2/2)

Example 8

Suppose a specific study region is known to have 2400 wolves, of which 1100 have already been tagged.

2. Approximate the probability in part (1) using the binomial distribution.

Discrete Uniform RVs

Discrete Uniform RVs

- **Scenario:** There are N possible outcomes, which are all equally likely.
- Shorthand: $X \sim \text{Uniform}(N)$

X	$X = \text{Outcome of interest, with } x = 1, 2, ..., N$
pmf	$p_X(x) = P(X = x) = \frac{1}{N} \text{ for } x = 1, 2, 3, ..., N$
$\text{E}(X)$	$\text{E}(X) = \frac{N+1}{2}$
$\text{Var}(X)$	$\text{Var}(X) = \frac{N^2 - 1}{12}$

What discrete uniform RVs have we seen already?

Example 9

Examples of discrete uniform RVs

- In R: every time we used `sample(1:N, ...)` without `prob`, we simulated a discrete uniform RV!

Learning Objectives

1. Distinguish between Bernoulli, Binomial, Geometric, Hypergeometric, Discrete Uniform, Negative Binomial, and Poisson distributions when reading a word problem.
2. Identify the variable and the parameters in a word problem, and state what the variable and parameters mean.
3. Use the formulas for the pmf/CDF, expected value, and variance to answer questions and find probabilities.